

joys 78.0 LAS for being closely related to English, whereas Arabic and Hindi, SOV languages, modestly reach 38.2 and 36.2 points using the SVO encoder. Although Chinese has SVO order, it is often seen as head-final while English is strong head-initial. Perhaps, this explains the poor performance for Chinese.

5.2 Impact of initialization

Initializing foreign embeddings is the backbone of our approach. A good initialization leads to better zero-shot transfer results and enables fast adaptation. To verify the importance of a good initialization, we train a $\text{RAMEN}_{\text{BASE}}+\text{BERT}$ with foreign word-embeddings that are initialized randomly from $\mathcal{N}(0, 1/d^2)$. For a fair comparison, we use the same hyper-parameters in §3.3. Table 3 shows the results of XNLI and UD parsing of random initialization. In comparison to the initialization using aligned fastText vectors, random initialization decreases the zero-shot performance of $\text{RAMEN}_{\text{BASE}}$ by 10.3% for XNLI and 11.6 points for UD parsing on average. We also see that zero-shot parsing of SOV languages (Arabic and Hindi) suffers random initialization.

	⌘	fr	vi	zh	ru	ar	hi	avg
XNLI	rnd	66.9	67.1	65.7	59.1	50.8	48.5	59.7
	scr	71.1	58.0	62.8	63.8	62.1	47.5	60.8
	vec	75.2	71.8	70.5	71.1	68.5	62.8	70.0
UD	rnd	71.0	33.2	25.5	60.3	19.7	13.4	37.2
	scr	63.8	17.0	13.6	61.0	15.5	11.2	30.3
	vec	78.0	38.3	30.1	67.2	40.6	38.6	48.8

Table 3: Comparison between random initialization (rnd) of language specific parameters and initialization using aligned fastText vectors (vec) and bilingual BERT trained from scratch (scr) for 400 hours.

To highlight the efficiency of our transferring approach, we compare $\text{RAMEN}_{\text{BASE}}+\text{BERT}$ with a bilingual BERT (bBERT) trained from scratch (scr). For a fair comparison, we use the hyper-parameters described in section §3.3. All bBERT models are trained for 2,300,000 updates (400 GPU hours), which is more than sixteen times longer than RAMEN. We emphasize two important observations from Table 3. First, the randomly initialized foreign embeddings RAMEN performs on par with bBERT on XNLI task and significantly better than bBERT on UD parsing. This suggests that a large amount of linguistic knowledge can be transferred through reusing the pretrained BERT encoder. Secondly, with a careful initialization of

foreign embeddings, our $\text{RAMEN}_{\text{BASE}}+\text{BERT}$ outshines bBERT with even just 20,000 updates (3.5 GPU hours) as shown in Figure 2.

5.3 Are contextual representations from RAMEN also good for supervised parsing?

All the RAMEN models are built from English and tuned on English for zero-shot cross-lingual tasks. It is reasonable to expect RAMEN s do well in those tasks as we have shown in our experiments. *But are they also a good feature extractor for supervised tasks?* We offer a partial answer to this question by evaluating our model for supervised dependency parsing on UD datasets.

	fr	vi	zh	ru	ar	hi	avg
mBERT	92.1	62.2	85.1	93.1	83.6	91.3	84.6
$\text{RAMEN}_{\text{BASE}}$							
+ BERT	92.2	63.3	85.0	93.3	83.9	92.2	85.0
+ RoBERTa	92.8	65.3	86.0	93.7	84.9	92.5	85.9
$\text{RAMEN}_{\text{LARGE}}$							
+ BERT	92.8	64.7	86.2	93.9	84.9	92.0	85.7
+ RoBERTa	93.0	66.4	87.3	94.0	85.3	92.8	86.5

Table 4: Evaluation in supervised UD parsing. The scores are LAS.

We used train/dev/test splits provided in UD to train and evaluate our RAMEN -based parser. Table 4 summarizes the results (LAS) of our supervised parser. For a fair comparison, we choose mBERT as the baseline and all the RAMEN models are initialized from aligned fastText vectors. With the same architecture of 12 Transformer layers, $\text{RAMEN}_{\text{BASE}}+\text{BERT}$ performs competitive to mBERT and outshines mBERT by +1.1 points for Vietnamese. The best LAS results are obtained by $\text{RAMEN}_{\text{LARGE}}+\text{RoBERTa}$ with 24 Transformer layers. Overall, our results indicate the potential of using contextual representations from RAMEN for supervised tasks.

6 Conclusions

In this work, we have presented a simple and effective approach for rapidly building a bilingual LM under a limited computational budget. Using BERT as the starting point, we demonstrate that our approach performs better than mBERT on two cross-lingual zero-shot sentence classification and dependency parsing. We find that the performance of our bilingual LM, RAMEN , correlates with the performance of the original pretrained English models. We also find that RAMEN is also